THIRD YEAR (Computer Science)

FALL SEMESTER EXAMINATIONS 2024

Time : 3 Hours

Batch 2022

Dated : 11-DEC-24

Max Marks : 60

Dr. Abdul Karim Kazi

Theory of Automata and Formal Languages - CT-364

- Read the questions carefully.
- Attempt all questions.
- All questions carry equal marks.
- Show all steps in each answer along with complete tuple information.

Q.1 **Explain** which of the following regular expressions (R.E.) are equivalent (justify your answer) where $\Sigma=\{x\}$:

i. x^*

ii. $x(xx)^*$

iii. $(xx)^*$

iv. $(xx)^*(x+\varepsilon)$

(CLO-1)

Q.2 **Construct** a Moore's machine to produce residue modulo (known as modulus) "6" of binary numbers.

(CLO-2)

Q.3 **Construct** a Deterministic Finite Automaton (DFA) where $\Sigma=\{x, y, z\}$ such that fourth symbol from right hand side (R.H.S.) should be "y".

(CLO-2)

Q.4 **Construct** a Push-Down Automata (PDA) and also explain the logic of implementation for the following language where $\Sigma=\{s, t, u, v\}$:

$$L = \{ s^x t^y u^y v^x \mid x>=1 , y>=1\}$$

Also, show the parsing of the following strings:

i) sstttuuuvv

ii) ssssstuuvvvv.

(CLO-2)

Q.5 **Construct** a Turing Machine (TM) and also explain the logic of implementation for the following language where $\Sigma=\{x, y, \#\}$:

$$L = \{ \omega \# \omega^{reverse} \mid \omega \text{ belongs to } (x , y)^* \}$$

Also, show the parsing of the following strings:

i) xxyxyy#yyxyxx

ii) xxyxxx#yxxyxx

(CLO-2)

Q.6 **Transform** the following R.E. into minimal DFA where $\Sigma=\{x, y\}$:

$$(xyx + yxy)^*$$

(CLO-3)

CamScanner

SEAT NO. CT-21090

NED UNIVERSITY OF ENGINEERING & TECHNOLOGY

THIRD YEAR(Bachelor of Science in Computer Science & Information Technology)

FALL SEMESTER EXAMINATIONS 2023

Batch 2021

Dated : 29-JAN-24

Time : 3 Hours

Sir Kamran

Max Marks : 60

Theory of Automata and Formal Languages - CT-364

- Read the questions carefully.
- Attempt all questions.
- All questions carry equal marks.

Q1.a **Explain** Automata and its types with one examples of each? **[CLO-1: 5 Marks]**

Q1.b **List down** the Chomsky hierarchy of the languages w.r.t. Machines? **[CLO-1: 5 Marks]**

Q2.a **Explain** the concept of minimization of DFA and its importance? **[CLO-1: 5 Marks]**

Q2.b **Explain** the concept of ambiguity in the grammar with example? **[CLO-1: 5 Marks]**

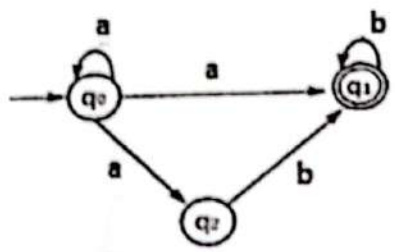
Q3.a **Construct** the ε-NFA for the regular expression: **010*1+ (1+0)*+101***? **[CLO-2: 5 Marks]**

Q3.b **Construct** Mealy machine that can count "**ab**" in the given string over {a, b}? **[CLO-2: 5 Marks]**

Q4.a **Construct** PDA for the language $\{a^{(2*m)} c^{(4*n)} d^n b^m \mid m,n \geq 1\}$? **[CLO-2: 5 Marks]**

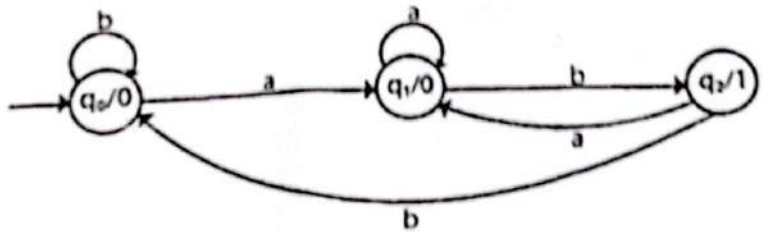
Q4.b **Construct** Turing machine for the language $L = \{ww^r \mid w \in \{0, 1\}\}$? **[CLO-2: 5 Marks]**

Q5.a **Transform** the following NFA to DFA over {a, b}? **[CLO-3: 5 Marks]**



W@ W^Y

Q5.b **Transform** the given Moore machine into Mealy machine? **[CLO-3: 5 Marks]**



Q6.a **Transform** the given CFG into CNF? **[CLO-3: 5 Marks]**

S -> aA | Aba, A -> AA | a, B -> AbA | bb

Q6.b **Transform** the mealy machine constructed in **Q3.b** into Moore machine? **[CLO-3: 5 Marks]**

SEAT NO. 32

NED UNIVERSITY OF ENGINEERING & TECHNOLOGY

THIRD YEAR(BACHELOR OF SCIENCE IN COMPUTER SCIENCE & INFORMATION TECHNOLOGY)

FALL SEMESTER EXAMINATIONS 2022

BATCH 2020

Dated:09-02-2023

Time: 3 Hours

Max.Marks:60

Theory of Automata and Formal Languages - CT-364

- Read the questions carefully.
- Attempt all questions.
- All questions carry equal marks.
- Show all steps in each answer along with complete tuple information.

Q#1 **Transform** the following regular expression into minimal DFA:

Table

$$(a+b)^*abb$$

Q#2 **Construct** a Moore's machine to produce residue modulo "5" of binary numbers.

Q#3 **Construct** a DFA where $\Sigma=\{a, b\}$ such that all following conditions should be met:

* Ending symbol must be "a".
* Length of the string should be greater than two but less than six.

Q#4 **Construct** a PDA and explain the logic for the following language:

$$L = \{ \ W @ W^R @@ \mid W \text{ belongs to } \{a, b\}^* \ \}$$

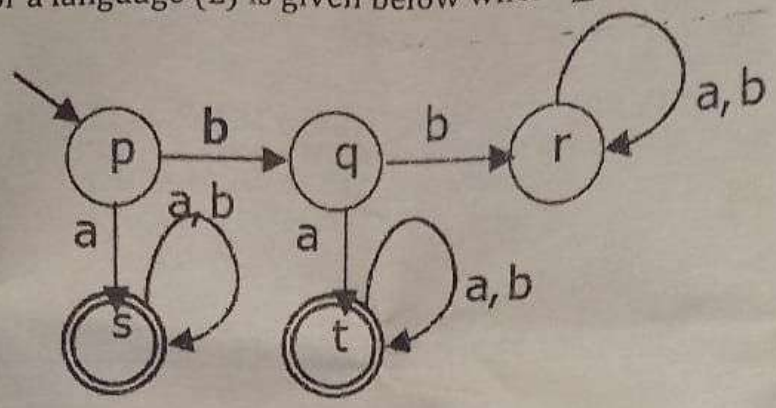
Also parse the following strings:

i) **ababbba@abbbaba@@**

ii) **bbaab@baabb@@**

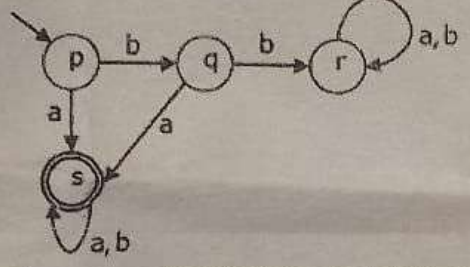
P.T.O

Q#5 A DFA for a language (L) is given below where $\Sigma=\{a, b\}$:

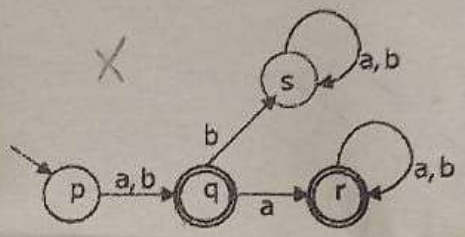


Explain which of the following finite state machine is a valid minimal DFA that accepts the same language as mentioned in above language (L)? (Justify reason for each following machine)

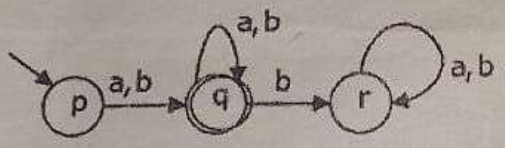
(A)



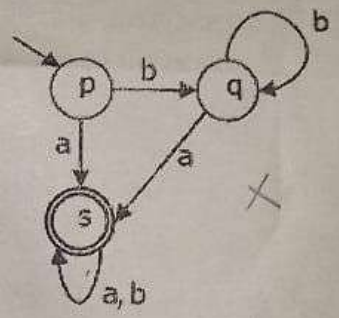
(B)



(C)



(D)



SEAT NO. _____

NED UNIVERSITY OF ENGINEERING & TECHNOLOGY

THIRD YEAR(BACHELOR OF SCIENCE IN
COMPUTER SCIENCE & INFORMATION TECHNOLOGY)
SPRING SEMESTER EXAMINATIONS 2021
BATCH 2018

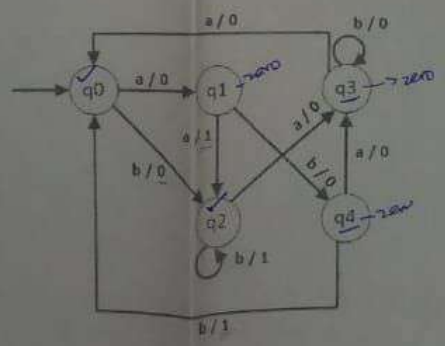
Time: 3 Hours

Dated:30-08-2021
Max.Marks:60

Theory of Automata and Formal Languages - CT-364

- Read the questions carefully.
- Attempt all questions.
- All questions carry equal marks.

Q1. Using the regular expression **a (b + c + d)* e + b (c + d + e)* f** where $\Sigma = \{a, b, c, d, e, f\}$; answer the following:

i) Analyze and explain the language it generates.
ii) Construct its NFA.
iii) Convert the NFA to DFA
iv) Minimize the DFA (if possible).

Q2. Convert the following Mealy machine (step by step) to its equivalent Moore machine: